HIGHER DIRECT IMAGES COMMUTE WITH FILTERED COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author marks the sheafification comparison 'some details omitted'.

Allowed standard prerequisites (editorial).

- delivered companion: colimits and cohomology. delivered companion lemma-colimits. Core colimit statement.
- Pushforward sheaf formula. Sheaves on Stacks, Lemma stacks-sheaves-lemma-pushforward-restriction. Reduction.

Source reference. Original reference 075G: sources/stacks-chow/tags.txt, source lines 9130–9130. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GQV: sources/stacks-chow/tags.txt, source lines 20875–20875. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. Let $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ be a filtered colimit of abelian sheaves on $\mathcal{X}$. Then for any $p \geq 0$ we have*

$$R^p f_* \mathcal{F} = \operatorname{colim} R^p f_* \mathcal{F}_i.$$

The same is true for abelian sheaves on $\mathcal{X}_{étale}$ taking higher direct images in the étale topology.

Proof. We will prove this for the fppf topology; the proof for the étale topology is the same. Recall that $R^i f_* \mathcal{F}$ is the sheaf on $\mathcal{Y}_{fppf}$ associated to the presheaf

$$(y : V \to \mathcal{Y}) \longmapsto H^i(V \times_{y,\mathcal{Y}} \mathcal{X}, \mathrm{pr}^{-1}\mathcal{F})$$

See Sheaves on Stacks, Lemma 075G. Recall that the colimit is the sheaf associated to the presheaf colimit. When V is affine, the fibre product $V \times_{\mathcal{Y}} \mathcal{X}$ is quasi-compact and quasi-separated. Hence we can apply Lemma 0GQV to $H^p(V \times_{\mathcal{Y}} \mathcal{X}, -)$ where V is affine. Since every V has an fppf covering by affine objects this proves the lemma. Some details omitted. □

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